Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT IV : Estimation and Testing of Hypothesis, Large Sample Tes

Syllabus: Estimation – parameters, statistics, sampling distribution, point estimation, null and alternative hypothesis, critical and acceptance regions, level of significance, two types of errors, power of test; large sample tests for proportions and means; confidence intervals.

I. Parameters, Statistics and Sampling Distribution

Definition:

Statistical inference: drawing conclusions about a population from a sample. It has two branches – ***estimation*** and ***testing of hypothesis***.

- ***Parameter***: numerical characteristic of the population (μ, σ, P) – fixed, usually unknown.
- ***Statistic***: function of sample observations ($\bar{x}$, s, p) – a random variable.
- ***Estimator***: statistic used to estimate a parameter; ***estimate***: its numerical value for a given sample.

Definition:

Sampling distribution: the probability distribution of a statistic computed from all possible samples of size n drawn from the population.

Standard error (S.E.): the standard deviation of the sampling distribution of a statistic.

Sampling distributions (large n)

Statistic	Mean	Standard error
Sample mean $\bar{x}$	μ	$\sigma/\sqrt{n}$
Sample proportion p	P	$\sqrt{(PQ/n)}$

Difference of means $\bar{x}1 - \bar{x}2$	$\mu1 - \mu2$	$\sqrt{(\sigma1^2/n1 + \sigma2^2/n2)}$
Difference of proportions $p1 - p2$	$P1 - P2$	$\sqrt{(P1Q1/n1 + P2Q2/n2)}$

- Finite population (sampling without replacement): multiply S.E. by $\sqrt{[(N - n)/(N - 1)]}$.

Central Limit Theorem

Definition:

If $x1, \ldots, xn$ is a random sample from any population with mean μ and variance σ^2, then for large n ($n \geq 30$) the sample mean $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$.

- Hence $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$. This is why normal tests work for large samples.
- If the population itself is normal, $\bar{X}$ is exactly normal for every n.

Note:

Large sample: $n \geq 30$. Small sample: $n < 30$ (Unit V). If σ is unknown, use the sample s in place of σ for large n.

2. Point Estimation

Definition:

Point estimation gives a single value (a point) as the estimate of an unknown parameter, e.g. $\bar{X}$ for μ.

Properties of a good estimator

- **Unbiasedness:** $E(T) = \theta$. Example: $E(\bar{X}) = \mu$; $E(s^2) = \sigma^2$ with $s^2 = \Sigma(x - \bar{X})^2/(n - 1)$.
  - The estimator with divisor n has $E = [(n - 1)/n]\,\sigma^2$ and is biased.

- **Consistency:** T converges to θ as n → ∞ (i.e. E(T) → θ and Var(T) → 0).
- **Efficiency:** among unbiased estimators the one with the smallest variance is most efficient. Efficiency = Var(T2)/Var(T1).
  - Example: for a normal population $\bar{x}$ is more efficient than the sample median (Var(median) ≈ $\pi\sigma^2/2n$).
- **Sufficiency:** T contains all the information about θ in the sample (Fisher - Neyman factorisation).

Methods of point estimation

- **Method of moments:** equate population moments to sample moments.
- **Maximum likelihood (MLE):** choose θ that maximises L(θ) = f(x1, θ) f(x2, θ) ... f(xn, θ). Solve d log L/dθ = 0.

Population	MLE
Binomial B(n, p)	$\hat{p} = x/n$
Poisson(λ)	$\hat{\lambda} = \bar{x}$
Normal N(μ, σ^2)	$\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \Sigma(x - \bar{x})^2/n$
Exponential(θ)	$\hat{\theta} = 1/\bar{x}$

Example:

Find MLE of λ for Poisson.

$L = \Pi e^{-\lambda}\lambda^{xi}/xi! \Rightarrow \log L = -n\lambda + (\Sigma xi)\log\lambda - \Sigma\log xi!$

$d\log L/d\lambda = -n + \Sigma xi/\lambda = 0 \Rightarrow \hat{\lambda} = \Sigma xi/n = \bar{x}$

Example:

Sample 4, 6, 8, 10, 12 from a population. Give point estimates of μ and σ^2.

$\bar{x} = 40/5 = 8$. $s^2 = (16 + 4 + 0 + 4 + 16)/4 = 10$.

3. Confidence Interval Estimation

Definition:

A $(1-\alpha) \times 100$ % confidence interval is a random interval that contains the true parameter with probability $1-\alpha$.

- 95 % CI means: if we repeat sampling many times, about 95 % of such intervals contain the parameter.

Confidence level	α	Critical z (two-sided)
90 %	0.10	1.645
95 %	0.05	1.96
98 %	0.02	2.33
99 %	0.01	2.58

One-sample problems

Formula:

Mean (σ known or n large): $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$ (use s if σ unknown)

Formula:

Proportion: $p \pm z_{\alpha/2} \cdot \sqrt{(pq/n)}$

Two-sample problems

Formula:

Difference of means: $(\bar{x}1 - \bar{x}2) \pm z_{\alpha/2}\sqrt{(\sigma1^2/n1 + \sigma2^2/n2)}$

Formula:

Difference of proportions: $(p1 - p2) \pm z_{\alpha/2}\sqrt{(p1q1/n1 + p2q2/n2)}$

- Margin of error $E = z_{\alpha/2} \times$ S.E. Wider interval $\Leftrightarrow$ higher confidence or smaller n.

Formula:

Sample size for mean: $n = (z_{\alpha/2}\ \sigma / E)^2$ for proportion: $n = z^2\ pq / E^2$

Example:

A sample of 100 has mean 52 and SD 10. Find 95 % and 99 % confidence limits for μ.

S.E. = 10/√100 = 1. 95 %: 52 ± 1.96 = **(50.04, 53.96)**

99 %: 52 ± 2.58 = **(49.42, 54.58)**

Example:

In a sample of 500, 40 % favour a scheme. Find 95 % CI for the population proportion.

p = 0.4, S.E. = √(0.4 × 0.6/500) = 0.0219. Limits: 0.4 ± 1.96 × 0.0219 = 0.4 ± 0.0429 = **(0.357, 0.443)**

Example:

Samples: n1 = 50, $\bar{X}1$ = 82, s1 = 8; n2 = 60, $\bar{X}2$ = 78, s2 = 6. Find 95 % CI for μ1 − μ2.

S.E. = √(64/50 + 36/60) = √1.88 = 1.371. 4 ± 1.96 × 1.371 = 4 ± 2.69 = **(1.31, 6.69)**

Interval does not contain 0, so the means differ significantly at 5 %.

Example:

How large a sample is needed to estimate a mean within 3 units with 95 % confidence if σ = 15?

$n = (1.96 \times 15/3)^2 = (9.8)^2 = 96.04 \Rightarrow \mathbf{n = 97}$

Example:

Sample sizes n1 = 400, n2 = 500; p1 = 0.15, p2 = 0.12. Find 95 % CI for P1 - P2.

S.E. = $\sqrt{(0.15 \times 0.85/400 + 0.12 \times 0.88/500)} = \sqrt{0.00053} = 0.0230$

$0.03 \pm 1.96 \times 0.0230 =$ **$(-0.015, 0.075)$** - contains 0 $\Rightarrow$ no significant difference.

4. Testing of Hypothesis - Basic Concepts

Definition:

Statistical hypothesis: a statement / assumption about a population parameter or distribution, which may or may not be true.

Definition:

Testing of hypothesis: a rule that decides, on the basis of sample data, whether to accept or reject the hypothesis.

Null and alternative hypotheses

- **Null hypothesis H0:** the hypothesis to be tested, stated as "no difference" or "no effect" (e.g. H0: $\mu = \mu_0$).
- **Alternative hypothesis H1 (Ha):** what we accept if H0 is rejected.
  - H1: $\mu \neq \mu_0$ → **two-tailed** test.
  - H1: $\mu > \mu_0$ (right-tailed) or $\mu < \mu_0$ (left-tailed) → **one-tailed** test.
- Simple hypothesis fixes all parameters; composite does not.

Test statistic, critical and acceptance regions

- **Test statistic:** a function of the sample whose distribution under H0 is known (e.g. Z).

- **Critical region (rejection region):** set of values of the test statistic for which H0 is rejected.
- **Acceptance region:** the remaining values where H0 is not rejected.
- Critical value: the boundary between the two regions.

Level of significance

Definition:

Level of significance α = P(rejecting H0 when H0 is true). Commonly 5 % or 1 %.

Level α	Two-tailed z	Right-tailed z	Left-tailed z
10 %	1.645	1.28	-1.28
5 %	1.96	1.645	-1.645
1 %	2.58	2.33	-2.33

Two types of errors

Decision	H0 true	H0 false
Reject H0	**Type I error** (α)	Correct (power)
Accept H0	Correct	**Type II error** (β)

- **Type I error:** rejecting H0 when it is true; P = α (producer's risk).
- **Type II error:** accepting H0 when it is false; P = β (consumer's risk).
- Reducing α increases β for a fixed sample size; both fall if n increases.

Power of a test

Formula:

Power = $1 - \beta$ = P(rejecting H0 when H1 is true)

- A good test has small α and large power. Most powerful test: highest power for a given α.

Example:

H0: $\mu = 100$ against H1: $\mu = 105$, $\sigma = 10$, $n = 25$, $\alpha = 0.05$ (right-tailed). Find β and power.

S.E. $= 10/5 = 2$. Critical value: $\bar{x} = 100 + 1.645 \times 2 = 103.29$. Reject H0 if $\bar{x} > 103.29$.

$\beta = P(\bar{x} < 103.29 \mid \mu = 105) = P(z < -0.855) =$ **0.196**; power = **0.804**

5. General Procedure for Testing

- Step 1: State H0 and H1 (decide one- or two-tailed).
- Step 2: Choose level of significance α.
- Step 3: Choose the test statistic and compute its value from the sample.
- Step 4: Find the critical value from the table.
- Step 5: Compare: if $|z_{cal}| > z_{tab}$ (in the critical region) reject H0; otherwise accept (do not reject) H0.
- Step 6: Write the conclusion in words.

Note:

Accepting H0 only means there is no sufficient evidence against it – it does not prove H0 is true.

Which test to use?

Situation	Test statistic
Single mean, n large	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Two means	$z = (\bar{x}_1 - \bar{x}_2)/S.E.$
Single proportion	$z = (p - P_0)/\sqrt{(P_0Q_0/n)}$
Two proportions	$z = (p_1 - p_2)/S.E.$ (pooled)
Small samples	t, F or χ^2 tests (Unit V)

6. Large Sample Tests – Single Mean

Formula:

H0: $\mu = \mu_0$. $z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ [use s if σ unknown] ~ N(0, 1)

- Conditions: $n \geq 30$ (or population normal with σ known).

Example:

A sample of 100 items has mean 52 and the population SD is 10. Test H0: $\mu = 50$ at 5 % level.

H0: $\mu = 50$, H1: $\mu \neq 50$ (two-tailed), $\alpha = 0.05$, $z_{tab} = 1.96$.

$z = (52 - 50)/(10/10) = 2$. $|z| = 2 > 1.96 \Rightarrow$ ***reject H0*** at 5 %.

At 1 % ($z_{tab} = 2.58$): $2 < 2.58 \Rightarrow$ accept H0.

Example:

A company claims bulbs last 1600 hours. A sample of 64 bulbs has mean 1570 hours and s = 120. Test at 5 % whether the claim is false (left-tailed).

H0: $\mu = 1600$, H1: $\mu < 1600$. $z = (1570 - 1600)/(120/8) = -30/15 = -2$.

$z_{tab} = -1.645$. Since $-2 < -1.645 \Rightarrow$ ***reject H0***; mean life is less than claimed.

Example:

A sample of 64 has mean 48.5 and s = 4. Test H0: $\mu = 50$ at 1 %.

$z = (48.5 - 50)/(4/8) = -3$. $|z| = 3 > 2.58 \Rightarrow$ ***reject H0***.

Example:

Average marks of 36 students is 70 with σ = 12. Is it consistent with population mean 75 at 5 %?

$z = (70 - 75)/(12/6) = -2.5$, $|z| > 1.96 \Rightarrow$ ***not consistent*** (reject H0).

7. Large Sample Tests – Difference of Means

Formula:

H0: $\mu_1 = \mu_2$. $z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

If $\sigma_1 = \sigma_2 = \sigma$ (common): $z = (\bar{x}_1 - \bar{x}_2) / [\sigma \sqrt{(1/n_1 + 1/n_2)}]$; if σ unknown replace σ^2 by s^2.

Example:

Two groups: $n_1 = 50$, $\bar{x}_1 = 82$, $s_1 = 8$ and $n_2 = 60$, $\bar{x}_2 = 78$, $s_2 = 6$. Is there a significant difference at 5 % and 1 %?

H0: $\mu_1 = \mu_2$, H1: $\mu_1 \neq \mu_2$. S.E. $= \sqrt{(64/50 + 36/60)} = \sqrt{1.88} = 1.371$.

$z = 4/1.371 = 2.92$. $2.92 > 1.96$ and $2.92 > 2.58 \Rightarrow$ ***reject H0*** at both levels – significant difference.

Example:

Mean income in city A ($n = 400$) is Rs 54 with SD 8; in city B ($n = 600$) Rs 52 with SD 6. Is A richer? (5 %, right-tailed)

S.E. $= \sqrt{(64/400 + 36/600)} = \sqrt{(0.16 + 0.06)} = 0.469$. $z = 2/0.469 = 4.26 > 1.645 \Rightarrow$ ***reject H0***: A has higher mean.

Example:

Two machines: n1 = n2 = 100, means 12.1 and 11.9, both SD = 1. Test equality at 5 %.

S.E. = $\sqrt{(1/100 + 1/100)} = 0.1414$. $Z = 0.2/0.1414 = 1.414 < 1.96 \Rightarrow$ ***accept H0***.

Note:

Steps never change - only the S.E. changes with the test.

8. Large Sample Tests - Single Proportion

Formula:

H0: $P = P0$. $Z = (p - P0) / \sqrt{(P0\ Q0 / n)}$, $p = x/n$, $Q0 = 1 - P0$

- Conditions: $nP0 > 5$ and $nQ0 > 5$ (large n).

Example:

In a sample of 400 items, 52 are defective. Test the claim that 10 % are defective at 5 %.

$p = 52/400 = 0.13$. S.E. $= \sqrt{(0.1 \times 0.9/400)} = 0.015$.

$Z = (0.13 - 0.10)/0.015 = 2 > 1.96 \Rightarrow$ ***reject H0***: claim not supported.

Example:

A coin is tossed 400 times and shows 216 heads. Is it fair? (5 %)

$p = 0.54$, $P0 = 0.5$, S.E. $= \sqrt{(0.25/400)} = 0.025$. $Z = 0.04/0.025 = 1.6 < 1.96$ $\Rightarrow$ ***accept H0*** (fair).

Example:

A manufacturer claims at least 95 % of goods are non-defective. Of 600 items, 570 are good. Test at 5 % (left-tailed).

$p = 570/600 = 0.95$. $Z = 0 \Rightarrow$ ***accept H0*** – claim holds.

Example:

In a survey of 1000 voters, 540 favour a candidate. Test H0: $P = 0.5$ against $P > 0.5$ at 1 %.

$p = 0.54$, S.E. $= \sqrt{(0.25/1000)} = 0.0158$. $Z = 0.04/0.0158 = 2.53 > 2.33 \Rightarrow$ ***reject H0***.

9. Large Sample Tests – Difference of Proportions

Formula:

H0: $P1 = P2$. $Z = (p1 - p2) / \sqrt{[\hat{P}\hat{Q}(1/n1 + 1/n2)]}$

Pooled proportion: $\hat{P} = (x1 + x2)/(n1 + n2) = (n1p1 + n2p2)/(n1 + n2)$, $\hat{Q} = 1 - \hat{P}$

Example:

In city A 60 of 400 smoke; in city B 60 of 500 smoke. Is the difference significant at 5 %?

$p1 = 0.15$, $p2 = 0.12$. $\hat{P} = 120/900 = 0.1333$, $\hat{Q} = 0.8667$.

S.E. $= \sqrt{(0.1333 \times 0.8667 \times (1/400 + 1/500))} = \sqrt{(0.11556 \times 0.0045)} = 0.0228$

$Z = 0.03/0.0228 = 1.32 < 1.96 \Rightarrow$ ***accept H0***: no significant difference.

Example:

Before a campaign 500 of 2000 favoured a product; after it 600 of 2000. Has the campaign been effective? (5 %, one-tailed)

$p1 = 0.25$, $p2 = 0.30$. $\hat{P} = 1100/4000 = 0.275$, $\hat{Q} = 0.725$. S.E. $= \sqrt{(0.275 \times 0.725 \times 2/2000)} = 0.01412$

$Z = (0.30 - 0.25)/0.01412 = 3.54 > 1.645 \Rightarrow$ ***reject H0***: campaign effective.

Example:

200 of 1000 tablets of type A and 170 of 1000 of type B were found defective. Test equality.

$\hat{P} = 370/2000 = 0.185$. S.E. $= \sqrt{(0.185 \times 0.815 \times 0.002)} = 0.01736$. $Z = 0.03/0.01736 = 1.73 < 1.96 \Rightarrow$ ***accept H0***.

10. Practice Problems with Answers

- 1) A sample of 49 gives mean 24 and SD 7. 95 % CI for μ? Ans: S.E. = 1, CI = (22.04, 25.96).
- 2) A sample proportion is 0.2 with n = 400. 99 % CI for P? Ans: S.E. = 0.02; $0.2 \pm 0.0516 = (0.148, 0.252)$.
- 3) Test H0: $\mu = 60$ if $n = 100$, $\bar{X} = 62$, $s = 8$ at 5 %. Ans: $Z = 2.5 > 1.96$, reject H0.
- 4) 45 of 100 patients recover with drug A, 60 of 120 with drug B. Test P1 = P2 at 5 %. Ans: $\hat{P} = 105/220 = 0.477$, S.E. = 0.0674, $Z = 0.05/0.0674 = 0.74$, accept.
- 5) Two samples: $n1 = 100$, $\bar{X}1 = 25$, $s1 = 3$; $n2 = 150$, $\bar{X}2 = 24$, $s2 = 4$. Test equality at 5 %. Ans: S.E. $= \sqrt{(0.09 + 0.1067)} = 0.4435$, $Z = 2.25$, reject.
- 6) Explain Type I and Type II errors with an example. Hint: court trial – convicting an innocent person is Type I; acquitting a guilty person is Type II.

- 7) Define level of significance and power of a test.

> **Note:**
> Writing tips: always state H0, H1, α, test statistic, calculated value, tabulated value, and the final conclusion in words.

II. More Worked Examples – Sampling and Estimation

Sampling distribution of the mean (illustration)

> **Example:**
> Population: 2, 4, 6 ($\mu = 4$, $\sigma^2 = 8/3$). All samples of size 2 with replacement: find the sampling distribution of $\bar{X}$.

Sample	(2,2)	(2,4)	(2,6)	(4,2)	(4,4)	(4,6)	(6,2)	(6,4)	(6,6)
$\bar{X}$	2	3	4	3	4	5	4	5	6

- Mean of all $\bar{X} = 36/9 = 4 = \mu$ ($\bar{X}$ is unbiased).
- Variance of $\bar{X} = (4 + 1 + 1 + 0 + 0 + 0 + 1 + 1 + 4)/9 = 12/9 = 4/3 = \sigma^2/n = (8/3)/2$ ✓.

Why $\bar{X}$ is unbiased and consistent

- $E(\bar{X}) = (1/n)\,\Sigma\, E(x_i) = n\mu/n = \mu$. $Var(\bar{X}) = n\sigma^2/n^2 = \sigma^2/n \to 0$ as $n \to \infty$.
- Therefore $\bar{X}$ is an unbiased and consistent estimator of μ.

Example:

A sample of 36 has mean 70 and σ = 12. Find 95 % and 90 % confidence limits for μ.

S.E. = 12/6 = 2. 95 %: 70 ± 3.92 = **(66.08, 73.92)**

90 %: 70 ± 1.645 × 2 = 70 ± 3.29 = **(66.71, 73.29)** (lower confidence ⇒ narrower interval)

Example:

Of 600 men, 300 and of 400 women, 325 prefer a brand. Give a test of equal preference.

$p1 = 0.5$, $p2 = 0.8125$, $\hat{P} = 625/1000 = 0.625$. S.E. = $\sqrt{(0.625 \times 0.375 \times (1/600 + 1/400))} = 0.03125$

$Z = 0.3125/0.03125 = 10 > 2.58 \Rightarrow$ ***reject H0*** - women's preference is much higher.

12. P-value and One-Tailed vs Two-Tailed Tests

Definition:

P-value: the probability, assuming H0 is true, of getting a test statistic at least as extreme as the one observed.

- Rule: if p-value < α reject H0; if p-value ≥ α accept H0.
- Two-tailed: $p = 2 \times P(Z > |z|)$. One-tailed: $p = P(Z > z)$ or $P(Z < z)$.

Example:

$Z = 2.0$ (two-tailed): $p = 2(1 - 0.9772) =$ **0.0456** $< 0.05 \Rightarrow$ reject at 5 % but not at 1 %.

Example:

$Z = 1.6$ (two-tailed): $p = 2(1 - 0.9452) =$ **0.1096** $> 0.05 \Rightarrow$ accept H0.

Choosing the tail

Wording in the problem	H1
"is different", "changed", "not equal"	two-tailed ($\neq$)
"greater", "higher", "more than", "improved"	right-tailed ($>$)
"less", "lower", "fewer", "reduced"	left-tailed ($<$)

- Critical region for two-tailed test is split: $\alpha/2$ in each tail. For one-tailed all α is in one tail.

Note:

One-tailed tests are more powerful in the stated direction but cannot detect change in the other direction.

13. Errors and Power – More Examples

Example:

To test H0: $p = 0.5$ against H1: $p = 0.75$ a coin is tossed 5 times. Reject H0 if 4 or more heads occur. Find α, β and power.

$\alpha = P(X \geq 4 \mid p = 0.5) = (5 + 1)/32 =$ **0.1875**

$\beta = P(X \leq 3 \mid p = 0.75) = 1 - [5(0.75)^4(0.25) + 0.75^5] = 1 - (0.3955 + 0.2373) =$ **0.3672**

Power $= 1 - \beta =$ **0.6328**

- Effect of changing things (other things fixed):
  - Increase n $\Rightarrow$ power increases, both α and β can be reduced.
  - Increase $\alpha \Rightarrow \beta$ decreases $\Rightarrow$ power increases.

- Larger true difference $|\mu_1 - \mu_0| \Rightarrow$ power increases.

- Neyman – Pearson lemma: for simple H0 vs simple H1, the most powerful test of size α rejects H0 when the likelihood ratio $L(\theta_1)/L(\theta_0)$ is large.

14. More Large Sample Tests – Examples

Example:

A die is thrown 600 times and a six appears 120 times. Is the die unbiased? (5 % and 1 %)

H0: P = 1/6. p = 0.2. S.E. = $\sqrt{(1/6 \times 5/6 / 600)}$ = 0.01521. Z = 0.0333/0.01521 = 2.19.

2.19 > 1.96 $\Rightarrow$ reject at 5 %; 2.19 < 2.58 $\Rightarrow$ accept at 1 %. ***Conclusion depends on the level.***

Example:

Heights of 400 men: mean 171.38 cm, population mean claimed 171.17 cm, σ = 3.3 cm. Test at 5 %.

Z = (171.38 – 171.17)/(3.3/20) = 0.21/0.165 = 1.27 < 1.96 $\Rightarrow$ ***accept H0.***

Example:

Two brands of tyres. Brand 1: n = 100, mean 1200 hours, s = 36. Brand 2: n = 100, mean 1180 hours, s = 40. Is brand 1 better? (5 %, one-tailed)

S.E. = $\sqrt{(36^2/100 + 40^2/100)}$ = $\sqrt{(12.96 + 16)}$ = 5.38. Z = 20/5.38 = 3.72 > 1.645 $\Rightarrow$ ***reject H0.***

Example:

A sample of 100 bolts has mean length 3.02 cm; the process mean is 3.00 cm with $\sigma = 0.1$ cm. Is the machine in order? (5 %)

$Z = 0.02/(0.1/10) = 2 > 1.96 \Rightarrow$ ***reject H0*** – machine out of adjustment.

Example:

20 % of a lot is claimed defective. A sample of 100 has 14 defective. Test H0: $P = 0.2$ against $P < 0.2$ at 5 %.

S.E. $= \sqrt{(0.2 \times 0.8/100)} = 0.04$. $Z = (0.14 - 0.2)/0.04 = -1.5 > -1.645 \Rightarrow$ ***accept H0***.

Note:

Compare $|Z|$ with the table value; for left-tailed tests compare the negative values carefully.

15. More Practice Problems

- 8) Sample of 200 gives mean 15 and SD 3. Find 98 % confidence limits. Ans: S.E. = 0.212, $15 \pm 2.33 \times 0.212 = (14.51, 15.49)$.
- 9) Can a sample of 100 with mean 49 and $s = 5$ come from a population with $\mu = 50$? (5 %) Ans: $Z = -2$, $|Z| > 1.96$, no.
- 10) 15 out of 60 boys and 25 out of 90 girls are vegetarian. Test equality of proportions at 5 %. Ans: $\hat{P} = 40/150 = 0.267$, S.E. $= \sqrt{(0.267 \times 0.733 \times (1/60 + 1/90))} = 0.0739$, $Z = (0.25 - 0.278)/0.0739 = -0.38$, accept.
- 11) Find the sample size to estimate a proportion within 0.05 at 95 % confidence when p is unknown (use $pq = 0.25$). Ans: $n = 1.96^2 \times 0.25/0.0025 = 384.16 \Rightarrow 385$.
- 12) Distinguish between parameter and statistic; estimate and estimator.

- 13) State the Central Limit Theorem and explain its use in large sample tests.

Quick Revision – Unit IV

- S.E. of $\bar{x} = \sigma/\sqrt{n}$; of $p = \sqrt{(PQ/n)}$.
- CI: estimate $\pm$ z $\times$ S.E. (z = 1.96 for 95 %, 2.58 for 99 %).
- α = P(Type I); β = P(Type II); power = 1 – β.
- Critical z (two-tailed): 1.645 (10 %), 1.96 (5 %), 2.58 (1 %).
- Reject H0 if $|z_{cal}| > z_{tab}$.
- Pooled $\hat{P} = (x1 + x2)/(n1 + n2)$ for the difference of proportions.